Darin Tsui

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Education

Georgia Institute of Technology, Ph.D. in Electrical and Computer Engineering — Aug 2023 – May 2028

- Advisor: Amirali Aghazadeh.

University of California San Diego, B.S. in Bioengineering — Sep 2019 – Jun 2023

Selected Publications

Darin Tsui, Kunal Talreja, Daniel Saeedi, and Amirali Aghazadeh. Protein circuit tracing via cross-layer transcoders. In *International Conference on Machine Learning (ICML)*, 2026.

Zain Shabeeb*, Daniel Saeedi*, **Darin Tsui***, Vida Jamali, and Amirali Aghazadeh. cryoSENSE: Compressive sensing enables high-throughput microscopy with sparse and generative priors on the protein cryo-EM image manifold. In *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.

Darin Tsui, Aryan Musharaf, Justin S. Kang, Yigit E. Erginbas, and Amirali Aghazadeh. SHAP zero explains biological sequence models with near-zero marginal cost for future queries. In *Conference on Neural Information Processing Systems (NeurIPS)*, 2025.

Thomas A. Walton, **Darin Tsui**, Aryan Musharaf, and Amirali Aghazadeh. SpecMER: Fast protein generation with K-mer guided speculative decoding. In *NeurIPS* (**Spotlight**), 2025.

Honors & Awards

AI for Science Program, Anthropic — 2026

- Awarded $50,000 in API credits toward circuit tracing in protein language models.

Steve W. Chaddick Fellowship, Georgia Tech — 2025

- Fellowship awarded for outstanding contributions in research.

NSF Graduate Research Fellowship — 2024 – 2027

- Fellowship recognizing potential for significant research contributions (top 14% of applicants).

Teaching Assistant's Choice Award (Mentor), UC San Diego — 2024

- Award recognizing the top-performing capstone project team out of 100 teams. Submissions were evaluated on technical depth, innovation, and team dynamic.

Course Enhancements & Improvement (CREATION) Award, Georgia Tech — 2023

- Award received for developing the course ECE 8803—*Generative and Geometric Deep Learning*.

Graduate Student Conference Scholarship Award, Conference on Information Storage and Processing Systems (ISPS) — 2023

- Award received for outstanding paper submission and oral presentation.

President's Fellowship, Georgia Tech — 2023

- Fellowship awarded to top 10% of Ph.D. applicants based on their research and scholarship.

Anushka Michailova Memorial Bioengineering Award, UC San Diego — 2023

- Award received for best capstone project out of 100 teams. Submissions were evaluated on technical depth, innovation, and research dissemination.

Research and Work Experience

AI ML and Information Research Group, Georgia Tech – Atlanta, GA — Jun 2023 – Present

- Advisor: Amirali Aghazadeh.
- Leveraged *mechanistic interpretability* frameworks, such as cross-layer transcoders and sparse autoencoders, for circuit discovery, protein design, and extracting biological mechanisms from protein language models.

- Extracted higher-order interactions in models 1000-fold quicker than state-of-the-art methods by developing algorithms at the intersection of *signal processing*, *coding theory*, and *combinatorics*.
- Developed principled *generative models* for high-throughput cryo-electron microscopy (cryo-EM), autoregressive protein generation, and variant effect prediction.

Generate:Biomedicines, Machine Learning Intern – Somerville, MA Jun 2026 – Aug 2026

- Advisors: Axel Levy and Max Baranov.
- Trained all-atom protein structure prediction foundation models to approximate protein dynamics.

Surgalign, Data Engineering Intern – San Diego, CA Jan 2023 – Aug 2023

- Manager: Weidong Luo.
- Interned at Surgalign, a medical device company focused on developing AI solutions in spinal surgery (now acquired by Xtant Medical and Augmetics).
- Automated MRI preprocessing pipelines and developed ground-truth spinal datasets for spinal surgery.

Talke Biomedical Device Lab, UC San Diego – San Diego, CA Dec 2021 – Aug 2023

- Advisor: Frank E. Talke.
- Designed low-cost, sub-millimeter surgical vision systems with OpenCV and Kalman filtering.

Integrated Systems Neuroengineering Lab, UC San Diego – San Diego, CA Jun 2022 – Jun 2023

- Advisor: Gert Cauwenberghs. Mentored by Ph.D. candidates Akshay Paul, Tyler Bodily, and Min Lee.
- Developed a GFET-based COVID-19 diagnostic test using an ML pipeline for signal classification, achieving 98.5% accuracy in protein detection.

Swartz Center for Computational Neuroscience, UC San Diego – San Diego, CA Mar 2021 – Jun 2023

- Advisor: Tzyy-Ping Jung. Mentored by Ph.D. candidate Chiyuan Chang.
- Conducted ML analysis of EEG data in Python/MATLAB, improving stress-level classification by 42.1% in correlation with physiological features.

Paesani Research Group, UC San Diego – San Diego, CA Jan 2020 – Sep 2021

- Advisor: Francesco Paesani. Mentored by postdoctoral scholar Vinícius Cruzeiro.
- Designed AMBER 20 molecular dynamics simulations on supercomputing clusters to model amino acid interactions via Python and Bash.

Publications

** Denotes equal contribution.*

Darin Tsui[*], Kunal Talreja[*], and Amirali Aghazadeh. Protein fitness landscapes are simpler under evolutionary distributions. *bioRxiv*, 2026.

Darin Tsui, Kunal Talreja, Daniel Saeedi, and Amirali Aghazadeh. Protein circuit tracing via cross-layer transcoders. In *International Conference on Machine Learning (ICML)*, 2026.

Darin Tsui, William Deinzer, Daniel Saeedi, and Amirali Aghazadeh. Circuit Tracing in Autoregressive Protein Language Models. In *Mechanistic Interpretability Workshop at ICML*, 2026.

Justin S. Kang, **Darin Tsui**, Yigit E. Erginbas, Landon Butler, Amirali Aghazadeh, and Kannan Ramchandran. Spectral sparsity: A unifying framework for scalable model interpretability using codes. *IEEE BITS the Information Theory Magazine*, 2026.

Zain Shabeeb[*], Daniel Saeedi[*], **Darin Tsui**[*], Vida Jamali, and Amirali Aghazadeh. cryoSENSE: Compressive sensing enables high-throughput microscopy with sparse and generative priors on the protein cryo-EM image manifold. In *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.

Darin Tsui, Aryan Musharaf, Justin S. Kang, Yigit E. Erginbas, and Amirali Aghazadeh. SHAP zero explains biological sequence models with near-zero marginal cost for future queries. In *Conference on Neural Information Processing Systems (NeurIPS)*, 2025.

Thomas A. Walton, **Darin Tsui**, Aryan Musharaf, and Amirali Aghazadeh. SpecMER: Fast protein generation with K-mer guided speculative decoding. In *NeurIPS* (**Spotlight**), 2025.

Darin Tsui, Kunal Talreja, and Amirali Aghazadeh. Sparse autoencoders for low-N protein function prediction and design. In *NeurIPS AI4Science Workshop*, 2025.

Thomas A. Walton, **Darin Tsui**, Lauren Fogel, Dustin J. E. Huard, Rafael S. Chagas, Raquel L. Lieberman, and Amirali Aghazadeh. GOLF: A generative AI framework for pathogenicity prediction of myocilin OLF variants. In *Machine Learning in Computational Biology*, 2025.

Darin Tsui[*], Kunal Talreja[*], and Amirali Aghazadeh. Efficient algorithm for sparse Fourier transform of generalized q-ary functions. In *IEEE Information Theory Workshop (ITW)*, 2025.

Darin Tsui, Kirsten Ramos, Capalina Melentyev, Ananya Rajan, Matthew Tam, Mitsuhiro Jo, Farshad Ahadian, and Frank E. Talke. A low-cost, open-source-based optical surgical navigation system using stereoscopic vision. *Microsystem Technologies*, 2025.

Darin Tsui and Amirali Aghazadeh. On recovering higher-order interactions from protein language models. In *International Conference on Learning Representations (ICLR) Generative and Experimental Perspectives for* Biomolecular Design (GEM) Workshop, 2024.

Chi-Yuan Chang, Chieh Hsu, Ying Choon Wu, Siwen Wang, **Darin Tsui** and Tzyy-Ping Jung. Online mental stress detection using frontal-channel EEG recordings in a classroom scenario. *arXiv preprint arXiv:2405.11394*, 2024.

Darin Tsui, Capalina Melentyev, Ananya Rajan, Rohan Kumar, and Frank E. Talke. An optical tracking approach to computer-assisted surgical navigation via stereoscopic vision. In *ISPS*, 2023.

Darin Tsui, Francisco Downey, Shreenithi Navaneethan, Akshay Paul, Tyler Bodily, Min Lee, Yuchen Xu, Ratnesh Lal, and Gert Cauwenberghs. A machine learning approach to COVID-19 detection via graphene field-effect-transistor (GFET). In *IEEE Engineering in Medicine and Biology Conference (EMBC)*, 2023.

Darin Tsui, Mitsuhiro Jo, Bryan Nguyen, Farshad Ahadian, and Frank E. Talke. Optical surgical navigation: a promising low-cost alternative. In *IEEE EMBC*, 2023.

Teaching Experience

Guest Lecturer, Georgia Tech – Atlanta, GA Nov 2023 – Nov 2025

- Delivered undergraduate and graduate-level lectures for *Generative and Geometric Deep Learning* (ECE 8803) and *Intro to Signal Processing* (ECE 2026).
- **Topics:** Explainability and mechanistic interpretability in AI [Link]; Generative and geometric deep learning in computational biology; IIR filters and feedback.

Teaching Assistant, Georgia Tech – Atlanta, GA Aug 2023 – Dec 2023

- Supported instruction for 300+ undergraduate and graduate students in *Advanced Programming Techniques* (ECE 4122/6122), focusing on C++ and parallel computing.
- Taught C++ development for lab assignments and mentored students through the completion of their final course projects. Held office hours for 10 hours a week.

Instructional Assistant, UC San Diego – San Diego, CA Dec 2020 – Mar 2021

- Led two lab sections of 30 students each in *Introduction to Bioengineering* (BENG 1), providing guidance on lab procedures and techniques.
- Taught and facilitated hands-on experiments across various bioengineering domains, including noninvasive cardiovascular sensing, medical image spine segmentation, and biomechanical analysis. Held office hours and lab sections for 10 hours a week.

Presentations

Protein circuit tracing via cross-layer transcoders, presented at the *Profluent ML Journal Club (Virtual)*, 2026.

Mechanistic interpretation of AI-designed proteins via cross-layer circuit tracing, presented at the *Suddath Symposium*, Atlanta, GA, 2026.

SHAP zero explains biological sequence models with near-zero marginal cost for future queries, presented at the *ML Student Seminar*, Atlanta, GA, 2025.

Surgical site localization with non-invasive skin markers for pain management, presented at the *Center for Memory and Recording Research (CMRR) Research Review*, San Diego, CA, 2023.

Bioelectronic COVID-19 detection via graphene-field-effect-transistor (GFET), presented at *Bioengineering Day at UC San Diego*, San Diego, CA, 2023.

Design of a surgical navigation system via positional tracking of fiducial markers, presented at the *Conference on Micromechatronics for Information and Precision Equipment*, San Diego, CA, 2022.

Science Communication

Darin Tsui. Global challenges in AI safety for biosecurity. *LessWrong*, 2026. [Link]

Darin Tsui. AI as biology's digital microscope. *LessWrong*, 2026. [Link]

Professional Development

BlueDot Impact, Participant — Virtual Jun 2026 – Oct 2026

- Selected for the Technical AI Safety and Biosecurity cohorts.

Global Challenges Project, Biosecurity Participant — Washington, D.C. Sep 2026

- Selected as one of 30 participants globally for the biosecurity cohort focusing on AI safety, biological risk, and policy evaluation.

Quadrant-i, Startup Launch Participant — Atlanta, GA Jun 2026 – Aug 2026

- Conducted market and technical research on commercializing mechanistic interpretability for protein design.

Service

Reviewer: 2026 ICML, NeurIPS, CVPR, ECCV, ICLR GEM; 2025 NeurIPS AI4Science; 2024 ICLR GEM.

Bioengineering Day Chair, Georgia Tech – Atlanta, GA Jan 2025 – Present

- Co-organized Bioengineering Day, a symposium highlighting interdisciplinary research with over 100 attendees.
- Coordinated logistics, speaker outreach, and scheduling for faculty and alumni talks, award presentations, and a rapid-fire thesis competition.

Capstone Project Advisor, UC San Diego – San Diego, CA Jul 2023 – Jun 2024

- Capstone project advisor for a bioengineering capstone project with Dr. Frank E. Talke.
- Mentored two undergraduates toward the design and development of a surgical navigation system with 3D tracking and augmented reality.
- Capstone project team won the *Teaching Assistant's Choice Award* at Bioengineering Day.

Founding Member, IEEE EMBS at UC San Diego – San Diego, CA Apr 2023 – Jul 2023

- Co-founded the IEEE Engineering in Medicine and Biology Society (EMBS) at UC San Diego with Dr. Gert Cauwenberghs and other Ph.D. students to increase awareness of bioengineering topics and research.
- Recruited 5 graduate students to mentor undergraduate students interested in getting involved in research.

President, IEEE at UC San Diego – San Diego, CA May 2022 – May 2023

- Oversaw operations for a 300+ student body. Partnered with the San Diego Supercomputing Center (SDSC) to co-found IEEE's Supercomputing Team.
- Hosted lab recruiting talks and technical seminars targeted towards first and second-years looking to get into research. Topics included research in artificial intelligence, robotics, and wearable sensors.
- Awarded the Outstanding IEEE Large Student Branch award for maintaining strong project activities and company relations during tenure.

Outreach Chair, IEEE at UC San Diego – San Diego, CA May 2021 – May 2022

- Taught 60 elementary and middle school students programming basics at STEM Fair.
- Hosted a robotics workshop in collaboration with Robolink, a San Diego robotics company, where we invited 50 high school students from low-income communities to work with Robolink representatives on coding and

designing their own aerial drones.

Project Team Lead, BMES at UC San Diego – San Diego, CA Oct 2021 – Jun 2022

- Led team of 20 undergraduates in designing a telesurgical prosthetic hand at UC San Diego's Biomedical Engineering Society (BMES). Hand designs were formulated and constructed based on biomechanics principles.
- Held workshops in CAD, Arduino, and circuit design. Implemented robotic motion in Arduino and interfaced linear actuators.

Committee Member, BMES at UC San Diego – San Diego, CA Sep 2019 – Jun 2023

- Helped organize Bioengineering Day, a day-long event celebrating all facets of bioengineering research and development, and Lab Expo, a research symposium aiming to get students involved in research and academia.

Patents

Amirali Aghazadeh, Darin Tsui, Kunal Talreja, Daniel Saeedi. (2026). *ProtoMech System*, (Application No. 63/981,521).

Amirali Aghazadeh, Daniel Saeedi, Vida Jamali, Darin Tsui, Zain Shabeeb. (2026). *Compressive Sensing Enables High-throughput Cryo-electron Microscopy Using Sparse and Generative Priors On the Protein cryo-EM Image Manifold*, (Application No. 63/977,630).

Mentoring

Bora Ak, Computer Engineering Undergraduate, Georgia Institute of Technology	2026
William Deinzer, CS Undergraduate, Georgia Institute of Technology *Research Intern at Intel*.	2026
Yash Choudhury, Computer Engineering BS, Georgia Institute of Technology	2026
Rishi Manimaran, CS BS/MS, Georgia Institute of Technology	2025 – 2026
Kunal Talreja, ECE Undergraduate, Georgia Institute of Technology *Research Intern at GTRI; President's Undergraduate Research Award*.	2024 – 2026
Aryan Musharaf, CS BS/MS, Georgia Institute of Technology *Engineer at Stealth Startup; Technical Staff at Hanover Park*.	2024 – 2025
Matthew Tam, Bioengineering Undergraduate, UC San Diego *Engineer at Briohealth Solutions*.	2023 – 2024
Kirsten Ramos, Bioengineering Undergraduate, UC San Diego *MEng Student at CSU Long Beach; Teaching Assistant's Choice Award*.	2023 – 2024
Eric Lee, Bioengineering Undergraduate, UC San Diego *MD Student at Loma Linda; Teaching Assistant's Choice Award*.	2023 – 2024
Capalina Melentyev, Bioengineering Undergraduate, UC San Diego *MS Student at Stanford*.	2022 – 2024
Rohan Kumar, Bioengineering Undergraduate, UC San Diego	2022 – 2023